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Taeun Kim

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EDUCATION

Emory University • *Bachelor of Science in Mathematics and Computer Science 4.0/4.0, 2x Dean's List* May 2027

- Coursework: Data Structures and Algorithms, Machine Learning, Database Systems, Computer Architecture, Linear Algebra, Multivariable Calculus

EXPERIENCE

Nell Hodgson Woodruff School of Nursing - MAIX Lab | AI Researcher Jun 2025 – Present

- Advanced multimodal **ECG** representation learning by integrating concept-based dictionaries into clinical report text embeddings, improving signal-report cross modal alignment accuracy by 9% over **MERL** approach.
- Applied **FAISS** for scalable disease concept retrieval, reducing retrieval latency by 65% over lexical methods.

Lablup | Research/ Software Engineering Intern Jun 2025 – Aug 2025

- Led GPU benchmarking and research automation, conducting vLLM performance analysis across Backend.AI's VM ecosystem to eliminate network latency variables, verified performance consistency, and built an auto-benchmarking pipeline with per-request token throughput metrics for internal.
- Enhanced Backend.AI FastTrack, an ML pipeline management service, by adding multi-selection and bulk deletion via GraphQL and React/Relay, and improving UI/UX with ReactFlow, dockerized workflows, and collaborative code reviews.
- Developed and deployed the PyCon 2025 booth event platform using Next.js, Tailwind, Shadcn UI, Google Apps Script, and AWS Amplify, supporting 228 participants and 1,193 submissions.

Sionic AI | ML Research Intern May 2024 – Jan 2025

- Researched **hard negative mining** (HNM) strategies to improve contrastive learning for the company's **RAG** pipeline.
- Boosted BGE-M3 fine-tuning by 4.87 points on Korean MTEB retrieval task with a positive-aware HNM strategy.
- Fine-tuned pretrained models with **PyTorch**, achieving an 85.73 BLEU in Korean grammar error correction, outperforming SOTA models GPT-4 (75.03) and HCX-3 (71.24); presented findings at PACLIC 38.

PROJECTS

Curaitor: AI-Powered Coursework Management Tool • [GitHub](#) • [YouTube](#) • [Devpost](#)

- A highly concurrent automated local file organizer and processor as Go backend, using the Actor Model with goroutines and channels, Gemini Go SDK, and mutex-protected caching with JSON-backed models for persistence.
- Expanded with Next.js/React frontend with Typescript and Tailwind CSS to display organized course materials, up-to-date quizzes and study guides, and interactive knowledge graphs across courses.

MBTI Dating Simulation App • [github/DolbonIn/MBTI-Date-Simulation](#)

- AI dating simulation app using Langflow based two stage **RAG** pipeline to generate MBTI-personalized conversations.
- Implemented dynamic, context aware dialogue, combining embedded personality traits with conversational memory.
- Technologies: Python, Streamlit, Langflow, Claude API, Cohere, Pinecone, Stable Diffusion

Good Hangul • [goodhangul.sionic.ai](#)

- On-premise Korean grammar correction platform powered by finetuned [nllb-200-ko-gec-3.3B](#) model hosted on Amazon S3, Flask API backend, PocketBase DB for collecting training data, and Next.js/React/TypeScript web interface.

LEADERSHIP AND COMMUNITY ENGAGEMENT

Emory Hacks | Emory Hacks Vice President and Co-Founder Aug 2023 – Present

- Founded Emory's first MLH-certified hackathon engaging 200+ participants; led the sponsorship team, securing \$20K+ from firms like Hyundai, JPMorgan, and Wolfram Alpha while building the official website with *Next.js* and *TypeScript*.

SKILLS

Languages: Go, Java, Python, JavaScript/Typescript, SQL, C, HTML, CSS, Shell Script

Frameworks/Libs: Pandas, PyTorch, React, Next.js, Node.js, Django, FastAPI, GraphQL, Nginx, PostgreSQL

Tools: Git, Linux, Docker, , AWS (S3, EC2, Amplify), GCP (Cloud Run/Build, Gemini), Jira, Gemini SDK